

ANGELOPOULOS & BATES (2022) – A GENTLE INTRODUCTION TO CP

Foundations and Trends in ML (arXiv:2107.07511, v6 Dec 2022). Monograph-length, hands-on tutorial; the **canonical practitioner reference**.

- Thesis: any black-box predictor $\hat{f}$ + an exchangeable calibration set + a chosen **nonconformity score** $s(x, y)$ yields finite-sample valid prediction sets, **no distributional assumptions**.
- Four-step recipe: (i) heuristic uncertainty $\hat{f}$, (ii) score s , (iii) empirical quantile $\hat{q}$, (iv) set $\mathcal{C}(x) = \{y : s(x, y) \leq \hat{q}\}$.
- Score design governs **adaptivity and set size**; the **coverage guarantee is automatic** regardless of choice.
- Pairs each method with **Python notebooks** — explicit engineering hook that drove adoption.
- Surveys the entire post-2005 CP landscape: APS/RAPS, CQR, conformal Bayes, group-balanced CP, conformal risk control, weighted CP, full/cross/jackknife+,

ANGELOPOULOS–BATES: THE RECIPE IN FORMULAS

- **Empirical quantile** on calibration scores $\{s_i = s(X_i, Y_i)\}_{i=1}^n$:

$$\hat{q} = \text{Quantile}\left(\{s_i\}; \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right)$$

Theorem (Marginal coverage)

If $(X_i, Y_i)_{i=1}^{n+1}$ are exchangeable, $\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \in \left[1 - \alpha, 1 - \alpha + \frac{1}{n+1}\right]$.

- Coverage is a **random variable** over calibration draws; its distribution is **Beta** $\left(\lceil (n+1)(1-\alpha) \rceil, n+1 - \lceil (n+1)(1-\alpha) \rceil\right)$. Rule of thumb: $n \approx 1,000$ to keep coverage within $\pm 1\%$.
- Score catalogue:
 - Classification: $s = 1 - \hat{f}_Y(x)$; APS = cumulative softmax mass; RAPS adds a regulariser to avoid huge sets on tail classes.

ANGELOPOULOS–BATES: EXTENSIONS AND ENGINEERING

- **Conformal Risk Control**: replace the indicator loss $\mathbb{1}\{Y \notin \mathcal{C}\}$ with any **bounded monotone** loss $L(\lambda)$; tune threshold λ so $\mathbb{E}[L] \leq \alpha$. Generalises coverage to FNR, FDR, miscoverage in segmentation masks.
- **Learn-Then-Test**: extends risk control to **non-monotone** risks via multiple-testing corrections on a grid of λ .
- Approximate conditional coverage:
 - **Class-conditional**: separate $\hat{q}$ per class label.
 - **Group-balanced**: separate $\hat{q}$ per protected attribute.
 - **Mondrian**: partition $\mathcal{X}$ taxonomically, calibrate per cell.
- **Weighted CP** (Tibshirani et al. 2019) handles known **covariate shift** via likelihood-ratio weights.
- Compute-statistics spectrum: **Full CP** (exact, $O(|\mathcal{Y}|)$ refits) $\rightsquigarrow$ **Cross-CP / CV+** (K -fold) $\rightsquigarrow$ **Jackknife+** (LOO, $1 - 2\alpha$ guarantee) $\rightsquigarrow$ **Split CP** (single fit, loses calibration data).

DIEULEVEUT & ZAFFRAN (2025) – HI! PARIS TUTORIAL (91 SLIDES)

Joint UAI/ICML tutorial, delivered at Hi! PARIS Summer School (École Polytechnique, July 2025).

Cleanest entry into the Zaffran research line.

- Pedagogical arc: motivation $\leadsto$ SCP for regression+classification $\leadsto$ CQR / APS $\leadsto$ coverage hierarchy (marginal / conditional / calibration-conditional) $\leadsto$ Full CP & Jackknife+ $\leadsto$ breaking exchangeability $\leadsto$ applied case studies.
- Distinguishing feature vs. Angelopoulos–Bates: explicit, mathematical treatment of when conformal coverage really holds — the conditional-coverage impossibility, PAC-style calibration-conditional bounds, and online-update guarantees.
- Proof architecture made explicit: quantile lemma (a population-quantile inequality on exchangeable scores) is shown as the single technical workhorse behind SCP, full CP, and weighted CP.
- Three featured Dieuleveut-group results: ACI for time series (Zaffran–Feron–Goude–Josse–Dieuleveut 2022); CP-MDA-Nested for missing values

DIEULEVEUT–ZAFFRAN: PROOF ARCHITECTURE & COVERAGE HIERARCHY

- **Three flavours of coverage**, weakest to strongest:
 1. **Marginal**: $\mathbb{P}(Y \in \mathcal{C}(X)) \geq 1 - \alpha$ (over joint draw of test point *and* calibration set).
 2. **Calibration-conditional** / PAC-style (Vovk 2012; Bian–Barber 2023): for fixed cal. set, $\mathbb{P}_{\text{test}}(Y \in \mathcal{C}(X)) \geq 1 - \alpha - \epsilon(n, \delta)$ w.p. $\geq 1 - \delta$.
 3. **Conditional**: $\mathbb{P}(Y \in \mathcal{C}(X) \mid X = x) \geq 1 - \alpha \forall x$.

Lemma (Vovk impossibility, paraphrased)

*Distribution-free conditional coverage at non-trivial level is **achievable only by sets of infinite Lebesgue measure**.*

- Hence the deck's two coping strategies: **partition + per-cell CP** (Mondrian / class-cond.), or **accept PAC slack** via calibration-conditional bounds.
- **Jackknife+** (Barber–Candès–Ramdas–Tibshirani 2021):

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}^{J+}(X_{n+1})) \geq 1 - 2\alpha$$

DIEULEVEUT–ZAFFRAN: BREAKING EXCHANGEABILITY & CASE STUDIES

- **Known covariate shift** $P_X \neq Q_X, P_{Y|X} = Q_{Y|X}$ (Tibshirani et al. 2019): reweight scores by likelihood ratio $w(x) = dQ_X(x)/dP_X(x)$; recover marginal coverage on Q at the cost of an effective-sample-size penalty.
- **Time series, online setting:**
 - **OSSCP**: window-based SCP (Zaffran 2022; Dutot–Tibshirani–Candès 2024).
 - **ACI** (Gibbs–Candès 2021): $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathbb{1}\{Y_t \notin \mathcal{C}_t\})$; gradient-style online update yields **long-run miscoverage** $\rightarrow \alpha$ for *any* adversarial sequence, distribution-free.
- **Applied case studies:**
 - Image-to-image regression for healthcare UQ (Angelopoulos et al. 2022): conformal masks around pixel-wise predictions.
 - French day-ahead electricity spot prices: ACI handles spike-driven distribution drift better than rolling SCP.
- **Recent frontier: CP-MDA-Nested** extends mask-conditional validity under MCAR/MAR

STOCKER, MALGORZEWICZ, FONTANA, BEN TAIEB (2025) – CP FOR TIME SERIES

arXiv:2511.13608 — the **time-series companion** to Angelopoulos–Bates.

- Diagnosis: time-series data violates **exchangeability** via two independent mechanisms:
 1. **Temporal dependence**: ARMA-style filtration; observations carry information about each other through time.
 2. **Distribution shift**: abrupt breaks, gradual drift, seasonality — the joint distribution itself moves.
- Self-contained re-derivation of finite-sample marginal *and* conditional coverage for SCP under **β -mixing** (Assumptions A1–A5, Theorem A.3.1); explicit proofs in appendix expand Oliveira et al. (2024).
- Formally distinguishes time-series non-exchangeability (**ordinal index + temporal dependence + drift**) from other non-exchangeability settings (covariate shift, panel data).
- Controlled simulation suite (AR(1), ARMA(1,1), mean-shift, GARCH(1,1)) comparing SCP, WCP (3 variants), ACI (3 step sizes), EnbPI (3 refresh rates), Block-SCP $\leadsto$ maps the **validity**

STOCKER ET AL.: THE FOUR-FAMILY TAXONOMY

Every modern CP-for-time-series method lives in exactly one of:

1. **Reweight calibration data** $\leadsto$ **WCP, Nex-CP**, learned-weight variants.
 - Mechanism: down-weight stale residuals via exponential or linear recency kernels.
2. **Refresh residual pool via OOB bootstrap** $\leadsto$ **EnbPI, SPCI, SPCI-T**.
 - Mechanism: bootstrap LOO residuals; slide a fresh window forward as new data arrive.
3. **Adapt target miscoverage online** $\leadsto$ **ACI, AgACI**, time-varying step sizes, **Conformal PID Control**.
 - Mechanism: gradient update on α_t **guarantees long-run miscoverage** regardless of data process.
4. **Randomise over blocks rather than points** $\leadsto$ **Block CP** (transductive + split variants).
 - Mechanism: shuffle contiguous chunks to restore exchangeability *between* blocks.

Taxonomy is mechanistic: the four operate on **four distinct objects** (calibration weights, residual pool, miscoverage target, block ordering). Hybrid methods compose families.

STOCKER ET AL.: SIMULATION RESULTS & PRACTITIONER DEFAULTS

Headline empirical findings from the controlled suite:

- **Stationary β -mixing** (AR(1), ARMA(1,1)): vanilla SCP is **valid and efficient**; WCP / ACI / EnbPI match coverage at varying widths and compute costs.
- **Abrupt mean shift**: SCP, Block-SCP, WCP-window **all fail to cover**; ACI, EnbPI, WCP-exp, WCP-linear **self-correct** via their explicit recency mechanisms.
- **Non-obvious failure**: Block-SCP under-covers *even on stationary β -mixing data* — a counter-intuitive cost of randomising blocks rather than points.

Practitioner defaults (from the paper's discussion):

- Stationary $\leadsto$ **SCP**.
- Cheap shift-robustness $\leadsto$ **WCP-exp** or **WCP-linear**.
- Active feedback / unknown drift $\leadsto$ **ACI** (or AgACI for aggregation).
- Heteroskedasticity matters, compute available $\leadsto$ **EnbPI** / **SPCI**.

Frontier: hybrid methods (ACI + SPCI conditional quantiles, learned WCP weights);

XU & XIE (2023) – ENBPI (IEEE TPAMI)

IEEE Trans. Pattern Analysis and Machine Intelligence (arXiv:2010.09107). The canonical reference for the “refresh residual pool” family in Stocker’s taxonomy.

- Problem: time-series data isn’t exchangeable, so naive SCP loses its validity guarantee; ARIMA-style intervals require parametric assumptions; quantile regression needs heavy retraining.
- Core insight: a bootstrap ensemble already contains LOO information — for each i , the bootstrap models that didn’t see index i aggregate to a leave-one-out estimator $\hat{f}^{-i}$ for free.
- Result: no data split, no retraining at test time, and asymptotic conditional coverage under β -mixing — rare in CP, where conditional coverage is usually impossible at finite n .
- Authors: Chen Xu (Bloomberg, ex Georgia Tech), Yao Xie (Georgia Tech); follow-up paper SPCI (2022) replaces residual quantiles with a learned quantile-regression score.
- Slots into Stocker’s four-family taxonomy as the headline “residual-refresh” method; complements ACI (“adapt- α ”) for abrupt-shift recovery.

ENBPI: THE BOOTSTRAP-LOO TRICK

Algorithm (training data $(X_i, Y_i)_{i=1}^T$, ensemble size B):

1. Train B bootstrap point predictors $\{\hat{f}^{(b)}\}_{b=1}^B$.
2. For each i , aggregate (mean / median) **only** the bootstrap models whose training set excluded i :

$$\hat{f}^{-i}(x) = \frac{1}{|\{b: i \notin \mathcal{I}_b\}|} \sum_{b: i \notin \mathcal{I}_b} \hat{f}^{(b)}(x).$$

3. LOO residuals $\hat{e}_i = Y_i - \hat{f}^{-i}(X_i)$.
4. **Test-time interval** on a sliding window of recent residuals:

$$\hat{C}_t(X_t) = \hat{f}(X_t) + [\hat{Q}_\beta(\{\hat{e}\}), \hat{Q}_{1-\alpha+\beta}(\{\hat{e}\})]$$

where $\beta \in [0, \alpha]$ is chosen by **inner optimisation** to minimise width $\leadsto$ asymmetric intervals.

5. Model the residuals with a model $\hat{g}$ to get $\hat{e}_i$ and $\hat{e}_i$.

ENBPI: THEORY, EMPIRICS, LIMITATIONS

Theorem (Asymptotic coverage gap, informal)

*Under β -mixing of (X_t, Y_t) and consistency of the bootstrap LOO estimator, both the **marginal** and **conditional** coverage gaps of EnbPI converge to zero as $T \rightarrow \infty$. Also bounds the set difference between EnbPI's interval and the oracle (narrowest conditionally valid) interval.*

- Empirical benchmark across **solar/wind generation**, anomaly detection, greenhouse-gas concentration, appliance energy, Beijing air quality (PM2.5).
- Baselines beaten: **ARIMA**, **ICP**, **WeightedICP**, **QOOB**, **AdaptCI** (Gibbs–Candès), **J+aB**. EnbPI delivers **shorter intervals at nominal coverage**, robust to **missing data**, handles **change points** via sliding-window length.
- **Limitations**: degrades as β -mixing slows toward long memory; for **abrupt shift** pair with ACI (online α -update); for heteroskedasticity use the SPCI follow-up (learned quantile-regression score).
- Hybrid frontier: EnbPI residuals + ACI online correction: EnbPI + DB-ACI (Koukorinis 2026)

BAO, COLOMBO, MANOKHIN, CAO, LUO (2025) – CONFORMAL REGRESSION REVIEW

Proc. Conformal and Probabilistic Prediction and Applications (COPA), PMLR 266:1–23. **The practitioner decision aid** for univariate regression.

- Problem: the regression branch of the CP literature has **inconsistent notation** and a **proliferation of methods** (SCP, CQR, CHR, CTI, distributional CP, locally-weighted residuals, normalised SCP...) with no head-to-head reference.
- Contribution: **unified notation** + **categorisation** into score-design families + **empirical benchmark** on 12 real-world datasets.
- Stays focused on **univariate continuous targets** — deliberately narrow scope, which keeps comparisons apples-to-apples.
- Pairs with [164] (Angelopoulos–Bates) as foundations $\rightsquigarrow$ Bao tells you *which* regression score to pick once you've grasped the recipe.

Frames 2–3 below draw on the authors' categorisation; specific empirical numbers paraphrased

BAO ET AL.: THE METHODS COMPARED

8 methods examined, organised by the structure of the nonconformity score:

- **Residual-based** (symmetric intervals around $\hat{f}$):
 - **Split CP** (Papadopoulos et al. 2002; Lei et al. 2018): $s = |y - \hat{f}(x)|$.
 - **Normalised SCP**: $s = |y - \hat{f}(x)| / \hat{\sigma}(x)$; recovers asymmetry indirectly.
- **Quantile-based** (asymmetric, locally adaptive intervals):
 - **Conformal Quantile Regression (CQR)** (Romano et al. 2019): score = $\max(\hat{q}_{\alpha/2} - y, y - \hat{q}_{1-\alpha/2})$.
 - **Conformal Histogram Regression (CHR)** (Sesia–Romano 2021): histogram of conditional density $\rightsquigarrow$ score from cumulative mass.
 - **Conformalised Thresholded Intervals (CTI)**: thresholded conformal sets from full distributional output.
- **Distributional** (full predictive CDF):
 - **Distributional CP** (Chernozhukov et al. 2021): PIT-based score; intervals from any

BAO ET AL.: FINDINGS & METHOD SELECTION

Empirical findings on 12 real-world datasets:

- **No single winner** — ranking depends on dataset characteristics. The **marginal coverage guarantee is satisfied by all methods** (it has to be; that's the theorem); they differ on **interval width, conditional calibration, and compute**.
- **CQR / CHR / CTI** dominate on **heteroscedastic** and **multi-modal** data; their asymmetric intervals adapt to local conditional shape.
- **SCP and Normalised SCP** remain competitive on **roughly homoscedastic** data and have the **lowest compute and tuning footprint** — good defaults when in doubt.
- **Distributional CP** produces the best conditional calibration but **costs** an additional distributional regression model and substantially more calibration data.
- Trade-off triangle: **computational cost** $\leftrightarrow$ **interval width** $\leftrightarrow$ **calibration sample size** — the paper provides scatter plots that make this triangle empirical.

Method-selection heuristic synthesised by the authors: